Peaks or thresholds?

Non-Gaussian primordial black hole abundances for broad spectra and the pulsar-timing-array tension

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Abstract

The abundance of primordial black holes (PBHs) that accompanies an enhanced curvature spectrum is computed either with threshold statistics or with peaks theory. With local non-Gaussianity (NG), peaks theory had been applied only to monochromatic spectra or with the peak condition imposed on a Gaussian field. We construct peaks theory on the compaction function for broad spectra and local NG as the exact counterpart of threshold statistics, with the same threshold, window, transfer function and cut-off. The maxima of the non-Gaussian linear compaction $C_\ell = F'(\zeta_r)C_\mathrm{g}$ are the BBKS maxima of the Gaussian field $C_\mathrm{g} + \lambda\zeta_r$ with $\lambda = C_\ell F''/F'^2$, times a rank-one Hessian factor that we compute exactly to first order. An independent Monte Carlo and counts of maxima on 256^3 Gaussian random fields confirm the result; the Gaussian-peak approximation used so far miscounts peaks by up to 18%. The required amplitude ratio $R = A_*^\mathrm{peaks}/A_*^\mathrm{threshold}$ is 0.74–0.82 for $|f_\mathrm{NL}| \le 2$ and lognormal widths $0.1 \le \Delta \le 2$. The NG dependence cancels almost completely in R/R_Gauss, to within $\pm 4\%$ (16% for a strong ultra-slow-roll tail). Propagated to the NANOGrav 15-year data, peaks theory strengthens the exclusion of a scalar-induced gravitational-wave explanation by 0.2–0.8σ (0.4σ in the BBKS normalisation), for every NG template and spectral shape we consider. This is smaller than the effect of NG itself for $|f_\mathrm{NL}| \gtrsim 1$. We also add the primordial-NG corrections to the induced GW spectrum. We include all seven terms for local NG, validated against Adshead et al. They lower the significance by up to 0.26σ for $|f_\mathrm{NL}| \le 2$ and leave the gap between the statistics unchanged. The gap is comparable to three analysis choices that are rarely stated:

- when the transfer function acts in the collapse threshold;
- the normalisation of f_PBH (the expression in common use overestimates f_PBH by a factor of 3–13);
- the frequency range of the PTA fit.

We quantify all three.

1 Introduction

Pulsar timing arrays (PTAs) have found evidence for a nanohertz gravitational-wave background [1]. One cosmological interpretation is the gravitational waves induced at second order by a peak in the primordial curvature spectrum [3, 4, 5, 7, 2]. A peak large enough to produce the signal also forms PBHs when the perturbations re-enter the horizon. Requiring that PBHs are not overproduced disfavours the scenario at the 2–3σ level [8, 2, 9]. How strongly depends on the NG of the curvature perturbation and on how the PBH abundance is computed.

Two prescriptions are in use for the abundance:

- **Threshold statistics** counts the volume fraction in which the compaction function exceeds a threshold. It includes the non-linear relation between curvature and density and any local NG non-perturbatively [12, 13].

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- **Peaks theory** counts maxima [14]. It has been developed for Gaussian curvature with finite width [15, 16, 17] and for local NG with monochromatic spectra [18, 19, 20].

Several recent papers state that peaks theory with local NG for broad spectra is an open problem [13, 9, 10, 15, 17]. 2306.17149 [8] add, in a footnote, that peaks theory "provides slightly smaller amplitudes", so their threshold-statistics bounds are conservative.

This note fills that gap and quantifies the footnote.

- **Construction** (Sec. 2). We build peaks theory on the compaction function for broad spectra with local NG (quadratic, logarithmic or ultra-slow-roll, and curvaton templates). It is the exact counterpart of threshold statistics, so that the two differ only by the peak condition.
- **Main technical step.** We count maxima of the non-Gaussian compaction itself, a local function of two correlated Gaussian fields, rather than of its Gaussian component. This differs from the closest existing work [21, 22, 23].
- **The gap** (Sec. 4). We give the amplitude ratio $R(\Delta, \mathrm{NG})$ between the statistics.
- **PTA consequences** (Sec. 5). We propagate the ratio to the significance of the PBH-overproduction tension for NANOGrav 15, with every analysis choice stated and varied.
- **NG in the induced GWs** (Sec. 5.4). We add the primordial-NG corrections to the GW spectrum, which 2306.17149 neglect.
- **Two pitfalls.** The second appears widely in the literature.
  - The collapse threshold of 2306.17149 follows Musco et al.'s superhorizon smoothing, in effect without a transfer function. Their formula is written with the horizon-crossing transfer function, and read that way it changes the amplitude by 18–62%.
  - The f_{PBH} normalisation in common use is too large by a factor of 3–13.

2 Two statistics for the non-Gaussian compaction

2.1 Local NG and the compaction function

The curvature perturbation is $\zeta = F(\zeta_{\mathrm{g}})$ with ζ_{g} Gaussian and power spectrum $\mathcal{P}_\zeta$. We use three templates:

- quadratic, $F = \zeta_{\mathrm{g}} + \frac{3}{5} f_{\mathrm{NL}} \zeta_{\mathrm{g}}^2$;
- ultra-slow-roll (USR), $F = -\mu_* \ln(1 - \zeta_{\mathrm{g}}/\mu_*)$, which is $F = -\frac{2}{\beta}\ln(1 - \frac{\beta}{2}\zeta_{\mathrm{g}})$ of 2306.17149 with $\mu_* = 2/\beta$ and has $f_{\mathrm{NL}} = 5/(6\mu_*)$ at second order;
- the sudden-decay curvaton, $\zeta = \ln X(r_{\mathrm{dec}}, \zeta_{\mathrm{g}})$ [12].

At smoothing scale r (the scale that re-enters the horizon at formation, $k = 1/r$), define the Gaussian fields

$$C_{\mathrm{g}}(\boldsymbol{x}; r) = \int \frac{\mathrm{d}^3 k}{(2\pi)^3} e^{i\boldsymbol{k}\cdot\boldsymbol{x}} \tfrac{2\Phi}{3} (kr)^2 W(kr) T(kr)\, \zeta_{\mathrm{g}}(\boldsymbol{k}), \tag{1}$$

$$\zeta_r(\boldsymbol{x}; r) = \int \frac{\mathrm{d}^3 k}{(2\pi)^3} e^{i\boldsymbol{k}\cdot\boldsymbol{x}} W_s(kr) T(kr)\, \zeta_{\mathrm{g}}(\boldsymbol{k}), \tag{2}$$

with the top-hat $W(y) = 3(\sin y - y\cos y)/y^3$, the shell $W_s(y) = \sin y/y$, the radiation transfer function $T(y) = W(y/\sqrt{3})$ and $\Phi = 2/3$. All moments use a hard cut $kr < 4.49$, which the peak moments need for broad spectra; it changes threshold-statistics amplitudes by $\leq 2\%$. For a

spherical configuration $C_{\rm g} = -2\Phi r\zeta_{\rm g}'$, $\zeta_r = \zeta_{\rm g}(r)$, and the linear and full compaction of $\zeta = F(\zeta_{\rm g})$ are [12, 18]

$$C_\ell = F'(\zeta_r)\, C_{\rm g}, \qquad C = C_\ell - \frac{C_\ell^2}{4\Phi}. \tag{3}$$

A PBH of mass $M = \mathcal{K}(C - C_{\rm th})^\gamma M_H$ forms where $C > C_{\rm th}$ and $C_\ell < 2\Phi$ (type I). We use $\mathcal{K} = 3.3$ and $\gamma = 0.36$. The threshold $C_{\rm th} = \delta_c(q)$ follows the shape-dependent prescription of Musco et al. [24, 25], with the transfer function in the moments.

2.2 Threshold statistics

With $P_{\rm G}(c, z)$ the bivariate Gaussian of $(C_{\rm g}, \zeta_r)$, the mass fraction at one scale is [12]

$$\beta_{\rm th} = \int {\rm d}z \int_{C_{\ell,{\rm th}}}^{2\Phi} {\rm d}C_\ell \, \frac{P_{\rm G}(C_\ell/F'(z), z)}{|F'(z)|} \, \mathcal{K}(C - C_{\rm th})^\gamma, \tag{4}$$

where $C_{\ell,{\rm th}} = 2\Phi[1 - (1 - C_{\rm th}/\Phi)^{1/2}]$. Equation (4) includes the non-linear $C(C_\ell)$ and the full F.

2.3 Peaks of the Gaussian component (Gaussian-peak approximation)

The natural generalisation of earlier work is to count BBKS maxima of the Gaussian field $C_{\rm g}$, with ζ_r as a correlated scalar, and map them through F'. Where $F' < 0$, maxima of C_ℓ are minima of $C_{\rm g}$. This replaces the volume weight in (4) by

$$w = \frac{V\, G(m, s)}{(2\pi)^{3/2} R_*^3}, \qquad G(m, s) = \int_0^\infty {\rm d}x\, f(x)\, \mathcal{N}(x; m, s^2), \qquad V = \tfrac{4}{3}\pi r^3, \tag{5}$$

where:

- f is the BBKS function [14];
- m and s^2 are the conditional mean and variance of $x = -\nabla^2 C_{\rm g}/\sigma_2$ given $(C_{\rm g}, \zeta_r)$;
- $R_* = \sqrt{3}\sigma_1/\sigma_2$.

This is exact for $F' = {\rm const}$, but not in general: $\nabla C_\ell = F'\nabla C_{\rm g} + F'' C_{\rm g}\nabla\zeta_r$, so the maxima of C_ℓ are displaced from those of $C_{\rm g}$.

2.4 Peaks of the non-Gaussian compaction

The Kac–Rice density of maxima of C_ℓ at $(C_{\rm g}, \zeta_r) = (c, z)$ follows from three steps:

1. The gradient factorises:

$$\nabla C_\ell = F'(z)\, \nabla Y_\lambda, \qquad Y_\lambda \equiv C_{\rm g} + \lambda\zeta_r, \qquad \lambda = \frac{cF''(z)}{F'(z)}, \tag{6}$$

   with λ frozen at the point, so that Y_λ is a Gaussian field.

2. At a stationary point, $\nabla\nabla C_\ell = F'[\nabla\nabla Y_\lambda + \kappa\nabla\zeta_r\nabla\zeta_r^T]$, with $\kappa = -cF'(1/F')''$. The rank-one term vanishes identically for the USR template, since $1/F'$ is linear there.

3. Conditional on $\nabla Y_\lambda = 0$, the gradient $\nabla\zeta_r$ is Gaussian with variance $v = (\sigma_{1,cc}^2\sigma_{1,rr}^2 - \sigma_{1,cr}^4)/(3\sigma_{1,\lambda}^2)$. Here $\sigma_{n,AB}^2 = \int {\rm d}\ln k\, k^{2n} K_A K_B \mathcal{P}_\zeta$ and $\sigma_{n,\lambda}^2 = \sigma_{n,cc}^2 + 2\lambda\sigma_{n,cr}^2 + \lambda^2\sigma_{n,rr}^2$.

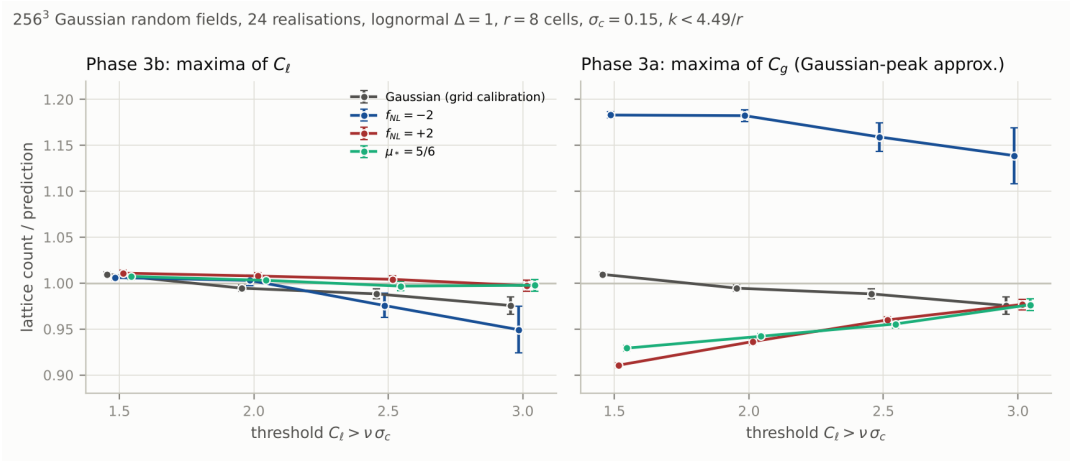


Figure 1: Maxima of $C_\ell = F'(\zeta_r)C_g$ counted on 256^3 Gaussian random fields, divided by the prediction of eq. (7) (left) and of the Gaussian-peak approximation (5) (right), as a function of the threshold in units of σ_c. Gray: Gaussian ζ, which calibrates the grid.

The determinant lemma then gives the peak density exactly to first order in $\tau = s\kappa\nu/\sigma_{2,\lambda}$ (Appendix A):

$$\boxed{n(c,z) = P_{\rm G}(c,z)\,\frac{G(s\,m_\lambda, s_\lambda)}{(2\pi)^{3/2}R_{*,\lambda}^3}\left[1 - \tau R_2(s\,m_\lambda, s_\lambda)\right], \qquad R_{*,\lambda} = \frac{\sqrt{3}\,\sigma_{1,\lambda}}{\sigma_{2,\lambda}},} \tag{7}$$

with $s = \operatorname{sign} F'$, m_λ and s_λ the conditional moments of $-\nabla^2 Y_\lambda/\sigma_{2,\lambda}$, and $R_2 = 27\int f_2\mathcal{N}/\int f\mathcal{N} \to 9/m$. The peak mass fraction is (4) with the integrand multiplied by $Vn/P_{\rm G}$. Equation (7) has three limits:

- it reduces to (5) for $\lambda = 0$ (Gaussian ζ or $F'' = 0$);
- it reduces to (5) for a monochromatic spectrum, where $Y_\lambda \propto \zeta_r$;
- it is finite as $F' \to 0$.

The rank-one factor is formally one power of the r.m.s. amplitude smaller, but $\kappa = -2\lambda^2/c$ for the quadratic template. It changes the local density by up to 36% at nodes with $f_{\rm NL} = -2$, $\Delta = 1$.

Checks.

- An independent Monte Carlo of the 20 Gaussian variables with the exact Hessian reproduces (7) to 3×10^{-4} at 15 of 16 test nodes, and to 0.85% at the most extreme one.
- We counted local maxima of C_ℓ on 256^3 Gaussian random fields (lognormal $\Delta = 1$, 24 realisations), with predictions from the lattices' own mode sums (Fig. 1). Eq. (7) follows the counts to about 1% beyond the grid trend seen in the Gaussian case. The Gaussian-peak approximation (5) is off by up to 18% for $f_{\rm NL} = -2$.

3 Abundance and its conventions

The abundance integrates the mass fraction over the horizon mass M_H at which $k = 1/r$ re-enters:

$$f_{\rm PBH} = \int \mathrm{d}\ln M_H\; w(M_H)\,\beta(M_H). \tag{8}$$

Three normalisations of w are in use (Appendix B):

- the $M_{\rm eq}$ form $w = (M_{\rm eq}/M_H)^{1/2}/\Omega_{\rm DM}$ [12, 17];
- eq. (14) of 2306.17149, $w = (M_H/M_\odot)^{-1/2}(g_*/106.75)^{3/4}(g_{*s}/106.75)^{-1}/(7.9\times10^{-10}\,\Omega_{\rm DM})$;
- the first-principles result from entropy conservation, which is the same expression with 1.0×10^{-8} in place of 7.9×10^{-10}.

The first two overestimate $f_{\rm PBH}$ by factors of 3–5 and 12.7 respectively. Since β depends exponentially on $1/A$, this lowers the amplitude for $f_{\rm PBH}=1$ by only $\ln(\text{factor})/|\ln\beta| \simeq 8\%$ and 12%. These factors cancel in R but not in the PTA tension.

The threshold prescription is a second, larger choice. The Musco et al. shape parameter q [24] is computed from the spectrum smoothed with the transfer function $T(k,\eta)$, and it matters when T acts:

- **Superhorizon.** Musco et al. apply it on superhorizon scales, $\hat r_m \simeq 10\,r_H$, where $T\simeq1$ except for modes with $k\hat r_m \gtrsim 20$. For every spectrum used here this agrees with no transfer function to ≤ 0.0005 in δ_c.
- **Horizon crossing.** Applied at horizon crossing, as in the variances, it smooths much more. For a lognormal of width $\Delta=1$ it gives $\delta_c=0.537$ instead of 0.458. For the broken power law of 2306.17149 ($\alpha,\beta,\gamma=4,3,1$) it gives 0.570 instead of 0.544.

2306.17149 write the horizon-crossing $T(k,r_m)$ in their eqs. (S4)–(S5). Their Fig. S3, however, follows the superhorizon prescription over its whole range (Appendix C). The two readings move A_* by 18% (BPL) and 62% (lognormal, $\Delta=1$) for every statistic, because the linear threshold $C_{\ell,\rm th}$ grows faster than $C_{\rm th}$. They hardly affect the ratio between the statistics: R changes by $\le 0.5\%$ for quadratic NG and by $\le 6\%$ for the curvaton and strong USR tails. Secs. 2–4 use the horizon-crossing threshold; Sec. 5 uses both.

Our implementation `ngpbh` was validated as follows:

- it reproduces the public code `pbhstat` [17] in its conventions to 0.017% in A_* for $\Delta=0.1$–2, for both statistics;
- it reproduces the broad-spectrum benchmark of Ferrante et al. (amplitude 9.09×10^{-3} vs 9.2×10^{-3});
- it reproduces their curvaton amplitudes to 1% [12].

The integration uses Gauss–Legendre nodes in C_ℓ that absorb the $(C-C_{\rm th})^\gamma$ cusp and an adaptive window in ζ_r, all in $\log\beta$. It is stable to 10^{-5} when every grid is doubled.

4 The gap between the statistics, with NG

Table 1 and Fig. 2 give $R = A_*^{\rm peaks}/A_*^{\rm threshold}$ for $f_{\rm PBH}=1$, with the same threshold, window, transfer function, cut, $\mathcal{K}$ and γ for both statistics.

- **Gaussian ζ:** $R=0.770$–0.789 in the BBKS normalisation with $V=\frac{4}{3}\pi r^3$.
- **Conventions:** R spans 0.75–0.89 across the normalisation conventions in use (Press–Schechter factor, peak volume, Jacobian). It is about 0.6–0.67 with the high-peak formula of Ref. [9] as used in Ref. [11]. These conventions cancel to better than 1% in the double ratio $R/R_{\rm Gauss}$.
- **The double ratio:** it stays within 0.958–1.036 for $|f_{\rm NL}|\le2$. The strongly non-perturbative USR tail $\mu_*=5/6$ reaches 1.05–1.16, the largest value at $\Delta=2$.

template	$\Delta = 0.1$	0.3	0.5	1	2
Gaussian	0.770	0.775	0.778	0.783	0.789
$f_{\rm NL} = -2$	0.738	0.744	0.769	0.794	0.810
$f_{\rm NL} = -1$	0.748	0.751	0.760	0.784	0.803
$f_{\rm NL} = +1$	0.788	0.792	0.796	0.803	0.811
$f_{\rm NL} = +2$	0.796	0.800	0.803	0.809	0.817
USR $\mu_* = 5/6$	0.808	0.816	0.825	0.851	0.914
USR $\mu_* = 5/2$	0.779	0.785	0.789	0.797	0.809

Table 1: $R = A_*^{\rm peaks}/A_*^{\rm threshold}$ with peaks of the non-Gaussian compaction (7), lognormal spectra of width Δ, $f_{\rm PBH} = 1$, $k_* = 10^7\,{\rm Mpc}^{-1}$.

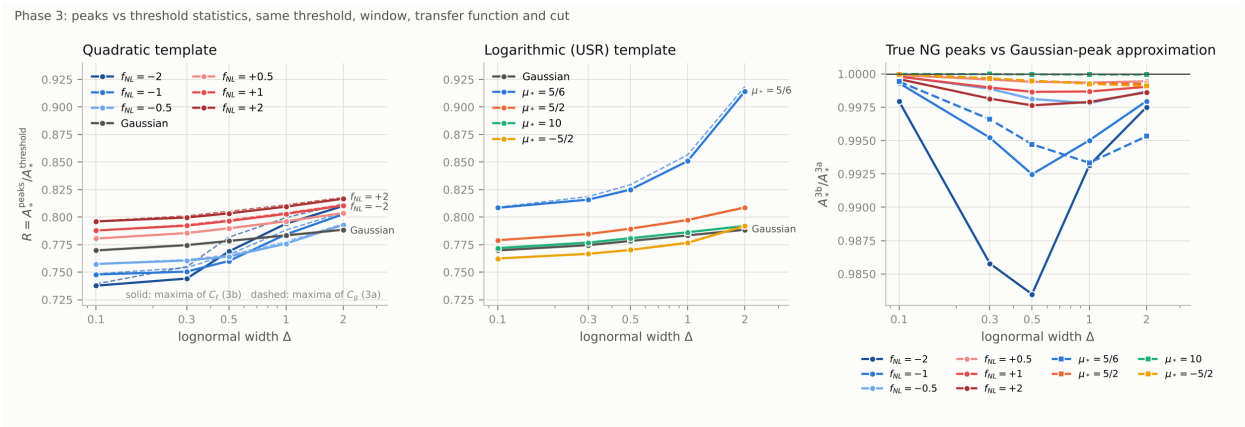


Figure 2: Left and middle: R for the quadratic and logarithmic templates; solid lines are peaks of C_ℓ (7), dashed lines are the Gaussian-peak approximation (5). Right: $A_*^{\rm 3b}/A_*^{\rm 3a}$.

- **True NG peaks vs the Gaussian-peak approximation:** the amplitudes differ by $A_*^{\rm 3b}/A_*^{\rm 3a} = 0.983$–$1.000$. The 18% miscount of peaks moves A_* by at most 1.7%, because $\delta \ln A_* \simeq \delta \ln n/|\ln\beta|$.

For comparison, NG moves the threshold-statistics amplitude itself by factors of 0.2–2 over the same grid.

5 Consequences for the PTA tension

5.1 Gravitational waves and the NANOGrav 15 likelihood

Induced GWs. For Gaussian ζ in radiation domination [5, 6, 8],

$$\Omega_{\rm GW,RD}(k) = \frac{1}{6}\int_1^\infty {\rm d}t \int_0^1 {\rm d}s \left[\frac{(t^2-1)(1-s^2)}{t^2-s^2}\right]^2 I^2_{t,s}\, \mathcal{P}_\zeta\Big(k\frac{t+s}{2}\Big)\mathcal{P}_\zeta\Big(k\frac{t-s}{2}\Big), \tag{9}$$

$$h^2\Omega_{\rm GW}(f) = h^2\Omega_r \frac{g_*}{g_*^0}\Big(\frac{g_{*s}}{g_{*s}^0}\Big)^{-4/3}\Omega_{\rm GW,RD}(k), \tag{10}$$

with $f = k/2\pi$, $h^2\Omega_r = 4.2\times10^{-5}$ and the Standard-Model $g_*(T)$ of [26]. The kernel is [5, 8]

$$I^2_{t,s} = \frac{288(s^2+t^2-6)^2}{(t^2-s^2)^6}\left[\frac{\pi^2}{4}(s^2+t^2-6)^2\Theta(t-\sqrt{3}) + \Big(t^2-s^2-\tfrac{1}{2}(s^2+t^2-6)\ln\Big|\frac{t^2-3}{3-s^2}\Big|\Big)^2\right]. \tag{11}$$

We rewrite the logarithmic bracket as $(3-s^2)\,g(e)$ with $e = (t^2-s^2)/(3-s^2)$ and $g = -e^3/12 - e^4/12 - \ldots$, so that the cancellation at $t, s \to 1$ is done analytically. We also split t at the resonance $t = \sqrt{3}$. Checks:

- a flat spectrum gives $0.8222436\,A^2$ [5];
- the Dirac closed form is recovered;
- direct two-dimensional integration agrees to 10^{-8}–10^{-12};
- the infrared k^3 tail matches its asymptotic form.

This is the spectrum for Gaussian ζ, as in [8]; Sec. 5.4 adds the primordial-NG corrections.

Likelihood. We use the NANOGrav 15-year free spectrum with Hellings–Downs correlations (the lowest 14 bins), represented by kernel density estimates of the per-bin posteriors in $\log_{10}\rho$ [27, 28]:

$$\ln L = \sum_j \ln p_j(\log_{10}\rho_j), \qquad \rho_j^2 = \frac{H_{100}^2\,h^2\Omega_{\rm GW}(f_j)}{8\pi^4 f_j^5 T}, \qquad T = 16.03\,\mathrm{yr}. \tag{12}$$

The KDEs use a compact kernel with bandwidths down to 0.004 dex. Outside the sampled range their density is floored, and their sparse tails are combs of single-sample spikes. We therefore regularise them by taking the maximum of each density and its 0.05-dex Gaussian blur (Appendix D). This leaves the bulk unchanged and moves every significance below by $\leq 0.14\sigma$.

Validation of the likelihood:

- for $h_c \propto f^{-2/3}$ it gives $A = 2.52^{+0.59}_{-0.61}\times10^{-15}$ (90%), against $2.4^{+0.7}_{-0.6}\times10^{-15}$ from NANOGrav [1];
- with a free slope it gives $\gamma = 3.07^{+0.63}_{-0.56}$, against 3.2 ± 0.6;
- for the lognormal SIGW fit with Δ marginalised it gives $\log_{10}A = -0.57^{+0.54}_{-0.84}$, against $-0.56^{+0.53}_{-0.74}$ in 2306.17149;
- for the broken power law with the shape marginalised it gives $\log_{10}A = -0.84^{+0.79}_{-0.74}$, against $-0.86^{+0.79}_{-0.65}$, and $\log_{10}k_* = 7.95$ against 7.92.

Significance. The posterior is taken in $(\log_{10}k_*, \log_{10}A)$, uniform on $[5,10]\times[-3,0]$, for the same spectral shape as the PBH line. The "allowed region" is $A < A_*(k_*)$, where $f_{\rm PBH}(A_*) = 1$. We quote two measures:

- σ_P, the one-sided Gaussian equivalent of the posterior mass in the allowed region;
- $\sigma_{\rm HPD}$, the two-sided equivalent of the highest-posterior-density level at which the region is first reached. This is the definition of 2306.17149: their $1\sigma/2\sigma/3\sigma$ contours lie at our 69/94/99% levels.

$\sigma_{\rm HPD}$ saturates on local features of a lumpy posterior ridge, so we lead with σ_P. Values above 4σ lie where the KDEs carry no information and are quoted only as ">4". The lines $A_*(k_*)$ for all k_* come from one set of β grids, because the moments depend on kr only.

5.2 Reproducing 2306.17149

Figure 3 compares our threshold-statistics bands with Fig. 3 of 2306.17149, read exactly from the vector paths of the arXiv source. Their setup reproduces their bands only with two of their conventions. The first is Musco et al.'s superhorizon threshold, which their Fig. S3 confirms (Appendix C). The second is their $f_{\rm PBH}$ normalisation. With both, the agreement for $k_* \geq 10^{7.5}\,\mathrm{Mpc}^{-1}$ is:

- Gaussian (non-linearities only): 1–6% in A;
- $f_{\rm NL} = -2$: within 6%;

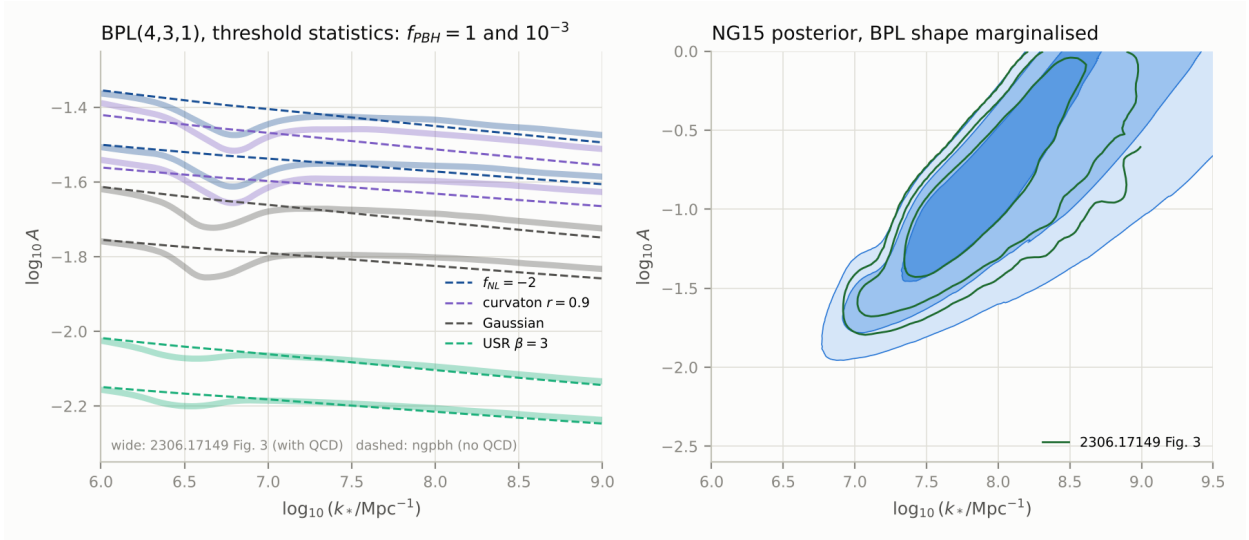


Figure 3: Left: threshold-statistics bands $f_{\rm PBH} = 1$ (upper) and 10^{-3} (lower) for BPL(4,3,1); wide lines are 2306.17149 Fig. 3 (with QCD), dashed lines are `ngpbh` (no QCD). Right: the NANOGrav 15 posterior with the BPL shape marginalised (fill: 68.3/95.4/99.7% HPD) and the contours of 2306.17149.

- USR with $\beta = 3$: within 2.3%;
- curvaton $r_{\rm dec} = 0.9$: 8–11%.

Their bands dip by up to 0.05 dex near $k_* \simeq 10^{6.5}$, where they include the QCD softening of the formation parameters and we do not. Putting their own lines on our posterior changes the significance by $\leq 0.07\sigma$, so the QCD omission does not matter, as they state.

5.3 Exclusion per statistic and NG template

Table 2 gives σ_P for threshold statistics and for peaks of the non-Gaussian compaction, in the conventions of 2306.17149. Figure 5 shows all templates.

- **The statistics shift.** Over the 81 cases with $\sigma_P \leq 4$, peaks theory adds $\Delta\sigma_P = +0.24$ to $+0.78$ (median $+0.42$). The amplitude shift itself is nearly uniform: -0.07 to -0.11 dex, and -0.05 for the strongest USR tail at $\Delta = 2$. It translates into different $\Delta\sigma$ depending on where the line meets the posterior.
- **Their setup.** With the shape-marginalised posterior and the BPL(4,3,1) lines, the Gaussian case goes from 2.3 to $2.8\sigma_P$ (1.4 to $1.9\sigma_{\rm HPD}$). $f_{\rm NL} = -2$ goes from 1.6 to $1.9\sigma_P$, and the curvaton from 1.7 to $2.0\sigma_P$. The USR $\beta = 3$ case, which they place beyond 3σ, gives $3.9\sigma_P$ with threshold statistics and > 4 with peaks.
- **True NG peaks vs the Gaussian-peak approximation.** They agree to $|\Delta\sigma_P| \leq 0.04$.
- **The NG shift.** Relative to the Gaussian case it runs from -1.2σ (negative $f_{\rm NL}$ and the curvaton, for narrow spectra) to well beyond $+2\sigma$ (positive $f_{\rm NL}$ and USR tails). For $\Delta = 2$ the sign of the $f_{\rm NL} < 0$ shift flips, because large negative $\zeta_{\rm g}$ also produce large compaction.

5.4 Primordial NG in the induced GWs

So far the GW spectrum is that of Gaussian ζ, as in 2306.17149. For the quadratic template $\zeta = \zeta_{\rm g} + F(\zeta_{\rm g}^2 - \langle\zeta_{\rm g}^2\rangle)$ with $F = 3f_{\rm NL}/5$, the four-point function of ζ ends at F^4. The spectrum

template	BPL(4,3,1)	LN $\Delta = 0.5$	LN $\Delta = 1$	LN $\Delta = 2$
Gaussian	$2.98 \to 3.62$	$3.02 \to 3.52$	$3.54 \to > 4$	$3.84 \to > 4$
$f_{\rm NL} = -2$	$1.87 \to 2.18$	$1.88 \to 2.31$	$3.43 \to 3.84$	> 4
$f_{\rm NL} = -1$	$1.98 \to 2.40$	$1.90 \to 2.36$	$3.05 \to 3.46$	$3.89 \to > 4$
$f_{\rm NL} = -0.5$	$2.25 \to 2.93$	$2.22 \to 2.89$	$3.09 \to 3.51$	$3.60 \to > 4$
$f_{\rm NL} = +0.5$	$3.67 \to > 4$	$3.60 \to > 4$	> 4	> 4
curvaton $r_{\rm dec} = 0.9$	$2.01 \to 2.44$	$2.00 \to 2.43$	$3.09 \to 3.49$	$3.91 \to > 4$
USR $\mu_* = 5/2$	$3.53 \to > 4$	$3.49 \to 3.91$	> 4	> 4
$f_{\rm NL} \geq +1$, USR $\mu_* \leq 5/6$	> 4	> 4	> 4	> 4

Table 2: Exclusion of the PBH-allowed region by NANOGrav 15, σ_P, threshold statistics $\to$ peaks of C_ℓ (7), $f_{\rm PBH} = 1$. Conventions of 2306.17149 (threshold without the transfer function, their $f_{\rm PBH}$ normalisation); fixed spectral shape in both the line and the posterior.

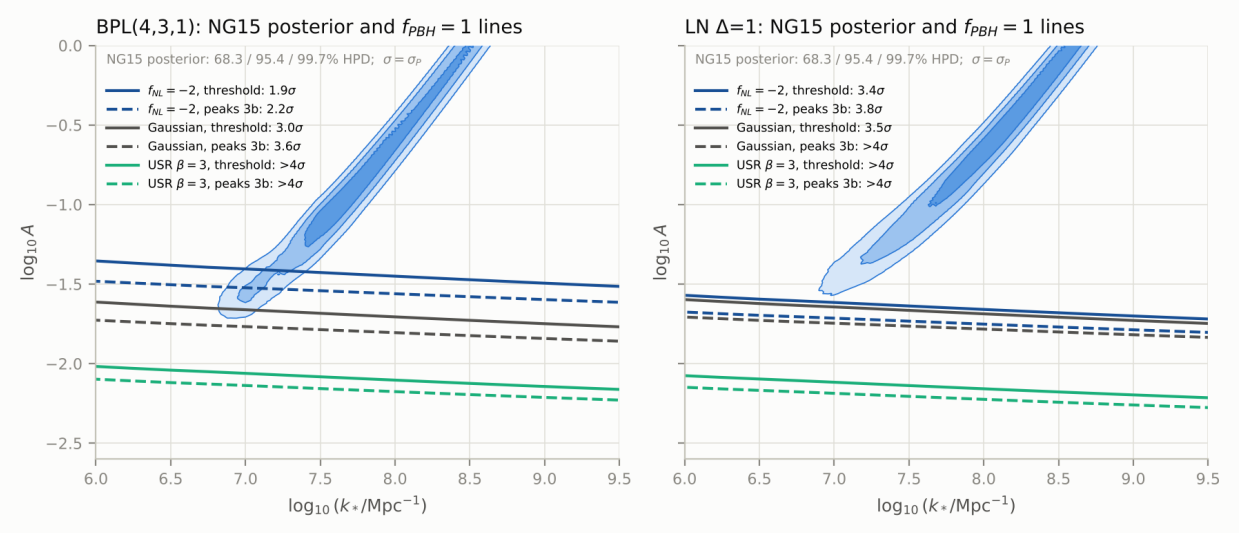


Figure 4: NANOGrav 15 posteriors (68.3/95.4/99.7% HPD) for BPL(4,3,1) and a lognormal of width 1, with the $f_{\rm PBH} = 1$ lines from threshold statistics (solid) and peaks of C_ℓ (dashed) for three NG templates. The legend gives σ_P.

then has seven terms [29]:

$$\Omega_{\rm GW,RD} = A^2 T_2 + F^2 A^3 \left(T_{\rm hyb} + T_C + T_Z\right) + F^4 A^4 \left(T_{\rm red} + T_{\rm pl} + T_{\rm np}\right). \tag{13}$$

For $\mathcal{P}_\zeta^{\rm g} = A\, s(k/k_*)$ each term depends on k/k_* only, and A is the amplitude of the Gaussian component, as in the PBH lines. All seven follow from

$$\Omega_X = \frac{k^3 c_X}{12\pi^2} \int \prod_{\rm loops} \frac{{\rm d}^3 q}{(2\pi)^3}\, {\rm Re}\left[z_1^2 \bar{z}_2^2\right] J_{12} \prod_i P(|\boldsymbol{m}_i|), \qquad J_{12} = \tfrac{1}{2}\left(I_{A1} I_{A2} + I_{B1} I_{B2}\right). \tag{14}$$

Here $z = (q_x + i q_y)/k$ is the transverse momentum of a GW vertex ($\boldsymbol{k} = k\hat{z}$). J_{12} is the oscillation average of the product of the late-time transfer functions $I = (I_A \sin x + I_B \cos x)/x$ of the two vertices [5]; J_{11} is the kernel above. The $\boldsymbol{m}_i$ are the momenta that carry a power spectrum, $P(m) = 2\pi^2 \mathcal{P}_\zeta^{\rm g}(m)/m^3$, and $c_X = 2$, 8, 16, 16, 8, 32, 16 for the Gaussian, hybrid, C, Z, reducible, planar and nonplanar terms. These reproduce the prefactors 2^5, $2^7 F^2$, $2^8 F^2$, $2^8 F^2$, $2^7 F^4$, $2^9 F^4$, $2^8 F^4$ of Ref. [29].

Numerics.

- **Disconnected terms.** They are the Gaussian functional of $P_{\rm g} + P_2$, with the one-loop spectrum $\mathcal{P}_\zeta^{(2)} = F^2 A^2 s_2$. We compute $T_{\rm hyb} = 2\,\Omega_{\rm bil}[s, s_2]$ and $T_{\rm red} = \Omega_{\rm G}[s_2]$ with the quadrature above.

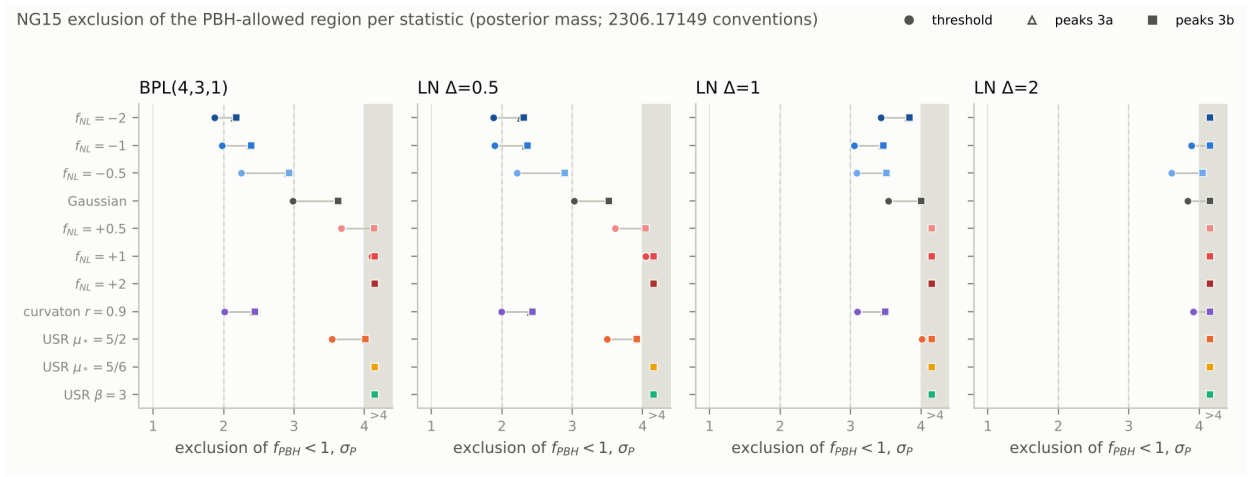


Figure 5: σ_P per NG template for threshold statistics (circles), the Gaussian-peak approximation (open triangles, hidden behind the squares) and peaks of C_ℓ (squares). Shaded: beyond 4σ.

- **Connected terms.** These are five- and eight-dimensional; we use randomised quasi-Monte Carlo over the spectral momenta. L of the $L+1$ momenta $\boldsymbol{m}_i$ are drawn from the spectrum, and the last follows from momentum conservation. The choices of which momentum is fixed are combined with the balance heuristic, so the weights stay bounded when a momentum becomes soft.

- **Variance for broad spectra.** Here pairs of nearly independent vertices dominate the variance: $\mathrm{Re}[z_1^2 \bar{z}_2^2] \propto \cos 2\phi$ averages to zero but scatters. We average each sample over rotations about $\boldsymbol{k}$ of one spectral momentum (for the planar term, of two, independently). The rotations preserve the measure and every weight, so the estimate stays unbiased; the oscillation cancels.

Checks.

- The Monte Carlo reproduces the quadrature of the Gaussian, hybrid and reducible terms: 72 comparisons, rms pull 0.97.

- All six NG terms agree with Fig. 4 of Ref. [29], read from the vector paths of the arXiv source (Fig. 6). For lognormal widths 1/5 and 1/10 the agreement is within 3% of each curve's maximum, including the sign changes of T_Z and the negative $T_{\rm np}$.

Other templates. For $\zeta = \zeta_{\rm g} + F\zeta_{\rm g}^2 + G\zeta_{\rm g}^3 + \ldots$, only F and G enter at $O(A^3)$. G enters through the power renormalisation $P \to (1+6G\sigma^2)P$, i.e. $+12G\sigma^2 A^2 T_2$; the $g_{\rm NL}$-type trispectrum vanishes under the azimuthal integral [30]. $O(A^4)$ would need G^2, F^2G and FH terms, so we truncate the curvaton ($F = -0.62$, $G = 0.32$ [31]) and the USR maps ($F = 1/2\mu_*$, $G = 1/3\mu_*^2$) at A^3. The quadratic template is exact at A^4.

Size. Relative to T_2 the corrections are large (Fig. 7). In the NANOGrav band of the posterior bulk, $k/k_* \simeq 0.02$–0.3, the ratio T_3/T_2 is 2–6. It grows with the width of the peak, because the one-loop spectrum collects the whole variance $\sigma^2 = A$ while the peak value of a broad lognormal is low. For $|f_{\rm NL}| = 2$, $\Omega_{\rm GW}$ at fixed A is therefore 10–40% larger near the PBH lines ($A \simeq 0.03$–0.05). In the posterior bulk it is more than doubled. The posterior median of $\log_{10} A$ moves down by 0.08–0.18 dex for $|f_{\rm NL}| = 2$, by 0.04–0.11 for the curvaton, and by ≤ 0.02 for $|f_{\rm NL}| = 0.5$.

Effect on the tension. The PBH-allowed region lies at small A, so the exclusion weakens (Table 3, Fig. 8). Over the cases with $\sigma_P \leq 4$:

- σ_P drops by 0.10–0.26 for $f_{\rm NL} = \pm 2$, by 0.03–0.20 for the curvaton, by ≤ 0.09 for $|f_{\rm NL}| = 1$, and by ≤ 0.03 for $|f_{\rm NL}| = 0.5$ and USR $\mu_* = 5/2$.

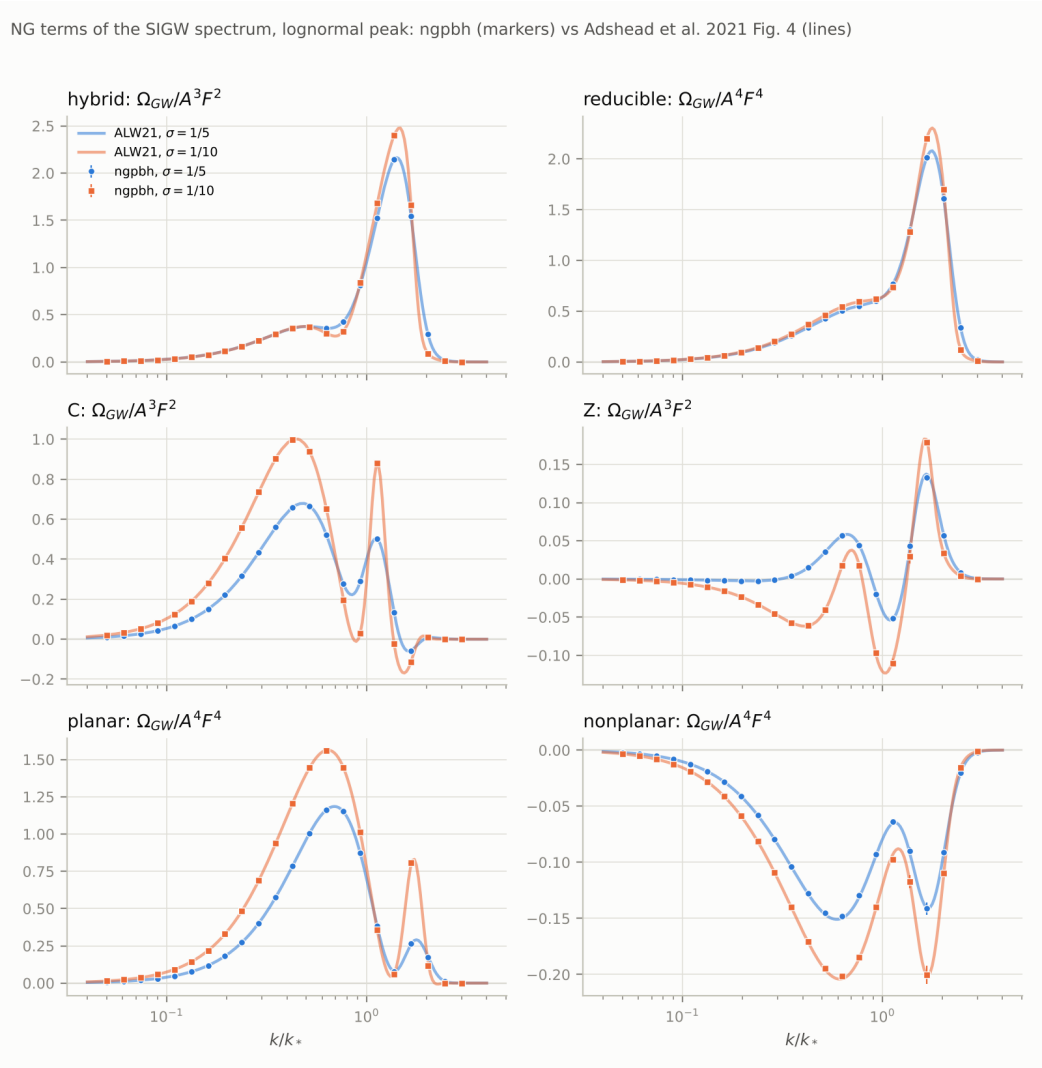


Figure 6: The six NG terms of the induced GW spectrum for lognormal peaks ($A = F = 1$): `ngpbh` (markers, Monte Carlo errors smaller than the markers) against Fig. 4 of Adshead, Lozanov & Weiner [29] (lines, read from the vector paths of the arXiv source).

- **The statistics shift is unchanged.** Threshold → peaks adds +0.23 to +0.67 (median +0.39), against +0.24 to +0.68 (median +0.42) with the Gaussian spectrum. None of the 66 pairs changes by more than ±0.10.
- **The truncation matters little.** Dropping the A^4 terms of the quadratic template, or adding the F^4A^4 terms to the others, moves σ_P by ≤ 0.02.

The GW part of the NG effect relieves the tension for either sign of $f_{\rm NL}$, by at most 0.26σ. The PBH part shifts σ_P by -1.2 to beyond $+2\sigma$.

5.5 Robustness

Table 4 varies one analysis choice at a time.

- **The statistics shift.** It survives every change: $+0.2$ to $+0.8\sigma$ across the peak-normalisation conventions in use, and $+0.8$ to $+1.3\sigma$ with the high-peak formula.
- **The absolute significance.** It depends mainly on the frequency range: 5, 14 or 30 bins move it by -1.5 to $+1\sigma$. The part of the posterior nearest the allowed region, $k_* \simeq 10^7\,\mathrm{Mpc}^{-1}$, places the GW peak at 10–20 nHz, and the bins above it bound it.

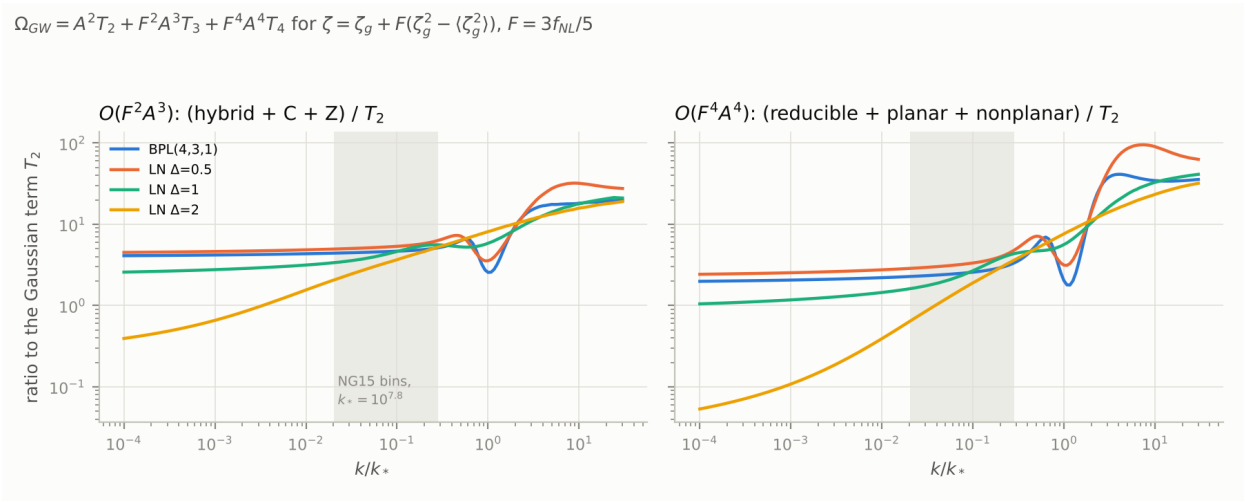


Figure 7: The NG corrections to the induced GW spectrum relative to the Gaussian term, per spectral shape. Shaded: the 14 NANOGrav bins for $k_* = 10^{7.8}\,\mathrm{Mpc}^{-1}$, near the posterior median.

template	GW spectrum	BPL(4,3,1)	LN $\Delta = 0.5$	LN $\Delta = 1$
$f_{\rm NL} = -2$	Gaussian	$1.87 \to 2.18$	$1.88 \to 2.31$	$3.43 \to 3.84$
	with NG	$1.73 \to 2.05$	$1.73 \to 2.11$	$3.25 \to 3.67$
$f_{\rm NL} = -1$	Gaussian	$1.98 \to 2.40$	$1.90 \to 2.36$	$3.05 \to 3.46$
	with NG	$1.94 \to 2.34$	$1.85 \to 2.30$	$2.99 \to 3.42$
$f_{\rm NL} = -0.5$	Gaussian	$2.25 \to 2.93$	$2.22 \to 2.89$	$3.09 \to 3.51$
	with NG	$2.24 \to 2.91$	$2.20 \to 2.87$	$3.07 \to 3.50$
curvaton $r_{\rm dec} = 0.9$	Gaussian	$2.01 \to 2.44$	$2.00 \to 2.43$	$3.09 \to 3.49$
	with NG	$1.92 \to 2.30$	$1.88 \to 2.26$	$2.95 \to 3.38$
USR $\mu_* = 5/2$	Gaussian	$3.53 \to > 4$	$3.49 \to 3.91$	> 4
	with NG	$3.51 \to 4.00$	$3.47 \to 3.90$	$3.99 \to > 4$

Table 3: σ_P for threshold statistics $\to$ peaks of C_ℓ with the Gaussian induced GW spectrum (as in Table 2) and with its primordial-NG corrections (each template with its own posterior). $f_{\rm PBH} = 1$, conventions of 2306.17149. The other templates are beyond 4σ with either spectrum.

- **The other conventions:**
  - the horizon-crossing transfer function in the threshold moves it by -0.2 to -1.0σ;
  - the entropy-conserving $f_{\rm PBH}$ normalisation by -0.15 to -0.4σ;
  - $f_{\rm PBH} = 10^{-3}$ instead of 1 by $+0.4$ to $+0.8\sigma$.
- **Fixed vs marginalised shape.** Fixing the shape raises the significance relative to marginalising over it (Gaussian BPL: 3.0 vs $2.3\sigma_P$). The fixed BPL(4,3,1) has a steep causality tail, which disfavours the low-k_* corner closest to the allowed region.

6 Discussion

What changes. The question we set out to answer is whether the gap between threshold statistics and peaks theory, once local NG is included for broad spectra, changes how strongly PBH overproduction disfavours a scalar-induced explanation of the PTA signal. It does, modestly and uniformly.

- Peaks theory lowers the overproduction line by about 0.1 dex for every NG template. This strengthens the exclusion by $\simeq 0.4\sigma$, or 0.2–0.8σ depending on the peak normalisation.

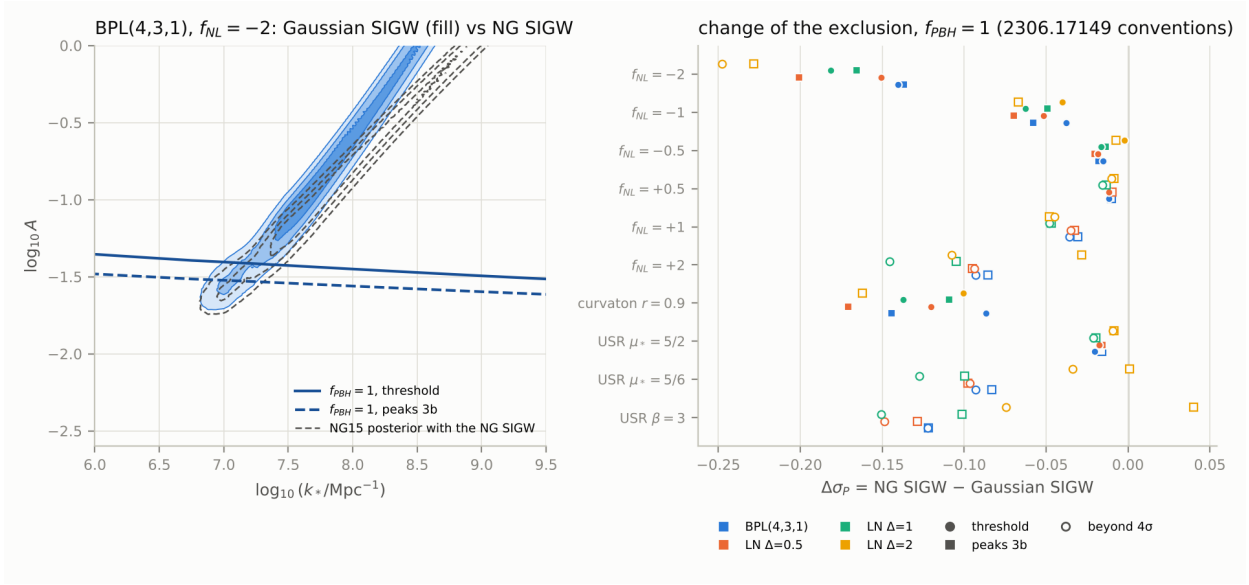


Figure 8: Left: the NANOGrav 15 posterior for BPL(4,3,1) with the Gaussian induced GW spectrum (fill) and with its NG corrections for $f_{\mathrm{NL}} = -2$ (dashed; 68.3/95.4/99.7% HPD), with the $f_{\mathrm{PBH}} = 1$ lines. Right: the change of σ_P from the NG corrections for every template and shape; open symbols lie beyond 4σ.

change from the reference	$\Delta\sigma_P$, threshold → peaks	max. change of σ_P
none (14 bins, KDE blur 0.05 dex)	+0.31 to +0.68	—
prior $\log_{10} k_* \in [6, 9]$, or $\log_{10} A \geq -4$	unchanged	0.00
posterior grid twice as fine	unchanged	0.00
KDE blur 0, 0.03 or 0.10 dex	+0.30 to +0.72	0.14
5 frequency bins (2–10 nHz)	+0.27 to +0.72	1.45
30 frequency bins (2–59 nHz)	+0.23 to +0.55	1.06
peak volume $(2\pi)^{3/2} r^3$	+0.40 to +0.81	0.14
Press–Schechter factor 2	+0.27 to +0.59	0.09
`pbhstat` v0.0.6 normalisation	+0.16 to +0.39	0.33
high-peak formula [9] (indicative)	+0.82 to +1.29	0.65

Table 4: Robustness of the PTA significances for BPL(4,3,1) and lognormals with $\Delta = 0.5$ and 1, over seven NG templates (cases with $\sigma_P \leq 4$). Reference prior: $\log_{10} k_* \in [5, 10]$, $\log_{10} A \in [-3, 0]$.

- The NG dependence of the gap cancels almost completely: the double ratio stays within ±4%. NG and the choice of statistics therefore add rather than interact.
- Negative f_{NL} and the curvaton remain the ways to relieve the tension, but peaks theory takes back 0.3–0.5σ of the relief.
- NG in the induced GWs adds power at fixed A. It relieves the tension by up to 0.26σ for $|f_{\mathrm{NL}}| \leq 2$ and does not change the gap between the statistics.
- None of the conclusions of 2306.17149 changes qualitatively. Their footnote is correct and now quantified.

What dominates. For an absolute statement about the PTA tension, the larger uncertainties lie elsewhere. The frequency range of the fit, the treatment of the collapse threshold and the f_{PBH} normalisation each move the significance by about as much as the choice of statistics, or more. All three are rarely stated. We recommend quoting them with any PBH bound on the PTA signal, together with the peak normalisation when peaks theory is used. On the theory side, our result

says that the Gaussian-peak approximation is adequate for the PTA question (to 0.04σ), although it miscounts peaks by up to 18%.

Limitations.

- As in the literature we compare with, we assume spherical symmetry for the compaction of a peak [18], use the mean profile (profile dispersion can matter for negative NG [22]), neglect QCD effects on the formation parameters and on the GW spectrum, truncate the NG series of the GW spectrum at A^3 for the curvaton and USR maps, use the free-spectrum refit rather than the full PTA likelihood, and ignore an astrophysical foreground. A supermassive-black-hole-binary component would relax every tension quoted here.
- The comparison between statistics is like-for-like, since all of these limitations affect both statistics in the same way.
- The absolute numbers carry the systematics of Table 4.

7 Conclusions

- **Construction.** We have constructed peaks theory with local NG for broad spectra on the compaction function, as the exact counterpart of threshold statistics. Maxima of the non-Gaussian compaction are maxima of a shifted Gaussian field, corrected by a rank-one term (eq. 7). Monte Carlo and lattice counts confirm this.
- **The gap.** Peaks require 18–26% smaller amplitudes than threshold statistics. This ratio is almost independent of NG for $|f_{\rm NL}| \leq 2$.
- **PTA consequence.** For NANOGrav 15 this strengthens the exclusion of a scalar-induced explanation by 0.2–0.8σ for every NG template. The effect is smaller than that of NG itself for $|f_{\rm NL}| \gtrsim 1$ and comparable to three analysis conventions, which we have quantified.
- **NG in the GWs.** The primordial-NG corrections to the induced GW spectrum, all seven terms, relieve the tension by up to 0.26σ for $|f_{\rm NL}| \leq 2$. They leave the gap between the statistics unchanged.
- **Code.** `ngpbh` implements every piece and reproduces published results: `pbhstat`, Ferrante et al., 2306.17149 and Adshead et al.

A Peaks of the non-Gaussian compaction: the rank-one factor

At a stationary point of C_ℓ, write $H_\lambda = \nabla\nabla Y_\lambda$ and $\boldsymbol{g} = \nabla\zeta_r$, with $\boldsymbol{g} \sim \mathcal{N}(0, v\mathbb{I})$ independent of H_λ. The exact density is

$$n(c,z) = P_{\rm G}(c,z)\Big(\frac{3}{2\pi\sigma_{1,\lambda}^2}\Big)^{3/2}\mathbb{E}\big[|\det(H_\lambda + \kappa\boldsymbol{g}\boldsymbol{g}^T)|\,\Theta_s\big], \tag{15}$$

where Θ_s requires negative- (positive-) definiteness for $s = +1$ (-1). For $H \prec 0$ the matrix determinant lemma gives $\det(-H-\kappa\boldsymbol{g}\boldsymbol{g}^T) = \det(-H)\,[1-\kappa\,\boldsymbol{g}^T(-H)^{-1}\boldsymbol{g}]$, which is linear in $\kappa|\boldsymbol{g}|^2$. Averaging over the direction and length of $\boldsymbol{g}$ gives $v\,\det(-H)\mathrm{tr}(-H)^{-1} = v\,e_2(-H)$. In the BBKS shape variables $e_2 = x^2/3-(3y^2+z^2)/3$, so $f_2(x) = \mathbb{E}[e_2\Theta|x]$ follows from the same two-dimensional integral as f; that integral reproduces BBKS's f to 10^{-10}. The interior part of the first-order factor is then exactly $1-\tau R_2$.

- For $\tau > 0$, maxima can only be lost, so $1-\tau R_2$ is a lower bound.
- For $\tau < 0$, configurations that become maxima add a positive $\mathcal{O}(\tau^2)$ term.

The Monte Carlo bounds the remainder at 0.85% at the most extreme node tested, a shift of 0.02% in A_*.

B The PBH abundance from entropy conservation

After formation at temperature T, the PBH number per entropy is conserved, so $\rho_{\mathrm{PBH},0} = \frac{3}{4}\beta\,(g_*/g_{*s})\,T\,s_0$. Here $s_0 = \frac{2\pi^2}{45} g_{*s}^0 T_0^3$ and β is the mass fraction at formation (including $\mathcal{K}(C - C_{\mathrm{th}})^\gamma$). With $M_H = 4\pi M_P^2/H$ and $H = \pi(g_*/90)^{1/2}T^2/M_P$ this gives

$$f_{\mathrm{PBH}} = \frac{1}{\Omega_{\mathrm{DM}}} \int \mathrm{d}\ln M_H \left(\frac{M_H}{M_\odot}\right)^{-1/2} \left(\frac{g_*}{106.75}\right)^{3/4} \left(\frac{g_{*s}}{106.75}\right)^{-1} \frac{\beta(M_H)}{1.0\times10^{-8}}, \tag{16}$$

in agreement with Carr et al. [32] to 2%. The constant 7.9×10^{-10} in eq. (14) of 2306.17149 makes f_{PBH} 12.7 times larger. The form $(M_{\mathrm{eq}}/M_H)^{1/2}\beta/\Omega_{\mathrm{DM}}$ drops the factor Ω_m and the heating of the radiation by g_{*s} changes after formation, and is 3–5 times larger. Both lower the amplitude for $f_{\mathrm{PBH}} = 1$ by $\ln(\text{factor})/|\ln\beta|$, i.e. 8–12% at $|\ln\beta| \simeq 21$.

C The collapse threshold of 2306.17149

Figure S3 of 2306.17149 plots δ_{th} from the Musco et al. prescription against the lognormal width and against the UV slope β of BPL(4,β,1). The figure is vector graphics in the arXiv source, so we read the curves exactly. We compare them with three readings of the prescription (Fig. 9):

- **No transfer function.** This reproduces their curves to ≤ 0.0002 for $0.1 \leq \Delta \leq 1$ and to ≤ 0.0016 for $1.9 \leq \beta \leq 3$.
- T **at** $\eta = \hat{r}_m/10$**.** This is Musco et al.'s superhorizon smoothing. It reproduces the curves to ≤ 0.0005 and ≤ 0.0012.
- T **at horizon crossing.** This is what the notation $T(k, r_m)$ of their eqs. (S4)–(S5) suggests. It is off by up to 0.08.

For $\beta < 2$ the integrals without T converge only conditionally, because the UV tail falls like $k^{-\beta}$. Their curve has a kink at $\beta \simeq 1.8$ and lies up to 0.007 below ours there. Their plotted α_s for the BPL is not the one behind their plotted δ_{th} through eq. (S6). For the lognormal both curves agree with ours. Their thresholds are therefore Musco et al.'s, and the conclusions of 2306.17149 do not depend on the notation.

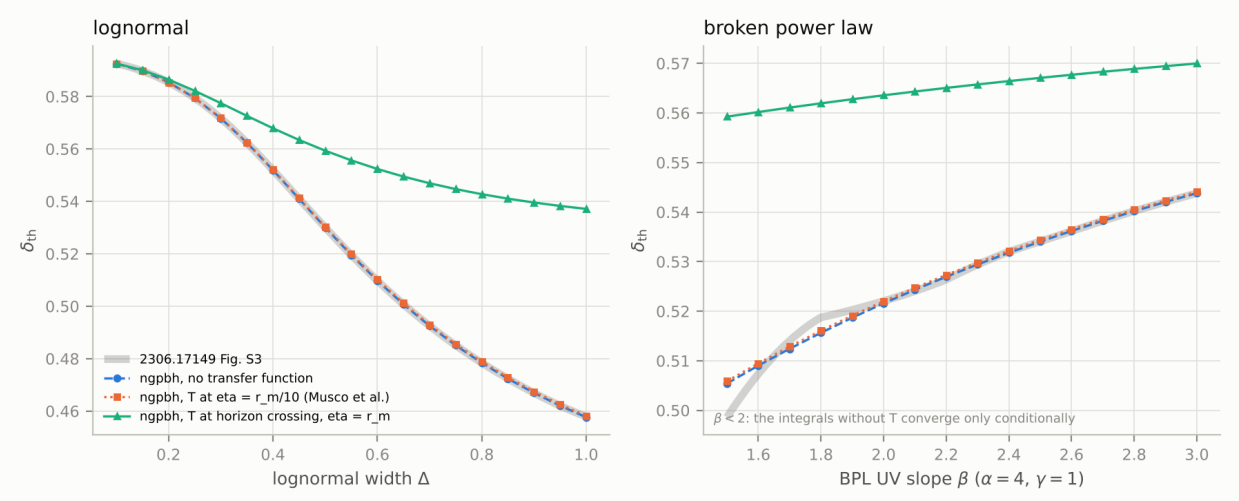


Figure 9: δ_{th} of Fig. S3 of 2306.17149 (wide gray), read from its vector paths, against `ngpbh` without the transfer function, with Musco et al.'s superhorizon smoothing ($\eta = \hat{r}_m/10$) and with the horizon-crossing transfer function.

D Free-spectrum density estimates

The per-bin densities distributed with PTArcade [28, 27] are Epanechnikov KDEs on a grid of 10^4 points in $\log_{10}\rho$. The kernel has compact support, so the density is exactly zero (stored as $\ln p = -36$) outside the range of the samples. Within the sparse tails it alternates between single-sample spikes and zero. The two most constraining bins (3.9 and 5.9 nHz) have almost no low-ρ tail. Tail probabilities beyond $\sim 2.5\sigma$ are therefore lumpy and cliff-like. We replace each density by the maximum of itself and its convolution with a 0.05-dex Gaussian: the bulk is unchanged and the holes are filled. Table 4 shows that the result changes by at most 0.14σ between no blur and 0.10 dex.

Code and data

`ngpbh` (MIT licence) is archived at Zenodo:

- this release: doi:10.5281/zenodo.23624507;
- all versions: doi:10.5281/zenodo.23624506.

It contains:

- the scripts that produce every table and figure here (`README.md`);
- 96 unit tests;
- the digitised Figs. 3 and S3 of 2306.17149 and Fig. 4 of Adshead et al.;
- the tables of the NG terms of the induced GW spectrum per spectral shape.

The NANOGrav 15 free-spectrum densities are those of PTArcade (Zenodo record 10495907) and are downloaded with an MD5 check.

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